Multi-isotope generator calculation

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Abstract

A brief explanation of the math behind solving arbitrarily complex isotope generator.

1 Introduction

Recent clinical developments [1] with the therapeutic agents based on Ac-225 prompted a vigorous search for practical methods to produce this isotope at scale. The clinical value of this isotope is estimated so high, that current market price of 1 patient dose of the isotoep alone currently goes as high as 10,000 USD, higher than any other isotope ever tried in humans. This high value started a frenzy of development and over a dozen of approaches were suggested employing many types of accelerators, nuclear reactors and isotopes generators. However, practical considerations of production at scale, with the right purity, in a distributed manner and robust enough to satisfy pharmaceutical manufacturing constrains eliminate most of the suggested methodologies. 3 methods currently supply most of the market: Th-229/Ac-225 generator, Ra-226(p,2n)Ac-225 reaction enabled by low energy cyclotron and Ra-226(γ,2n)Ac-225 process enabled by high energy eLINACs.

The last methodology is rapidly gaining popularity. The reason for this popularity is not the great advantages of the process, but rather lack of significant problems typically associated with the competitors. A common (p,2n) approach produces product contaminated with Ac-227; additionally potential failure of the radium target in this process can irreversibly contaminate the entire accelerator. Th-229/Ac-225 generator potentially is a very attractive technology, but the starting material is available commercially to only one company in the world, making this approach of no interest for any other developer. Photonuclear route also comes with significant problems. First, radium starting material is scantly available and hard to deal with in routine production. Second, eLINACs of sufficient power are unique instruments with less than a dozen of suitable units built in the entire world ever. Third, low conversion efficiency requires chemical processing of curies of Ra-226 to obtain meaningful quantities of Ac-225, necessitating large and complex GMP hot cell facilities. None of these problems, however are of fundamental nature, they all can be solved with the right investment and a capable team. As the old Jewish saying goes, "if a problem can be solved with money, it isn't a problem, it's an expense".

This review aims to provide a chemists perspective to the photonuclear production route.

This review focuses on the accelerator-driven experimental setups. While using photonuclear reaction to produce Ac-225 using a nuclear reactor [2] is an intriguing idea, the practical considerations of that approach are so vastly different from the accelerator-driven methods, that it is hard to draw meaningful comparison.

2 Two solutions to the same equations

Let's consider the most important isotope generator in nuclear medicine: Tc generator, that is still a staple of the clinical practice, used for millions upon millions of SPECT scans every year, all around the world. Mo-99 decays two ways: ˜87.6% to Tc-99m (branching ratio b) and ˜12.4% directly to (virtually) stable Tc-99, which does not factor into the generator considerations. From here, a conventional chemist's thinking can easily lead us to the following system of equations:

$$\begin{cases} \dfrac{d[^{99}Mo]}{dt} & = -\lambda_{Mo}[^{99}Mo] \\ \\ \dfrac{d[^{99m}Tc]}{dt} & = b\lambda_{Mo}[^{99}Mo] - \lambda_{Tc}[^{99m}Tc] \end{cases} \tag{1}$$

where decay constants are $\lambda_{Mo} = \ln 2/65.94h$, $\lambda_{Tc} = \ln 2/6.01h$ and branching ratio $b = 0.876$

What this is saying is that the rate of accumulation of ^{99}Mo is negatively proportional to its concentration, i.e. the more ^{99}Mo we have in solution the slower it accumulates, or what is the same, the faster it decays. This is essentially a definition of the radioactive decay. For the $^{99\mathrm{m}}$Tc the picture is more complex, but is still intuitive: its rate of accumulation is *negatively* proportional to its own concentration $[^{99m}Tc]$, but also *positively* proportional to the parent's concentration $[^{99}Mo]$. Also, let's not forget that the $[^{99}Mo]$ is scaled to the branching ratio in the second equation. Of course, the question that concerns the chemists the most is "What is in the beaker right now?", and the quest to answering this question can lead to two surprisingly divergent paths. The more intuitive one was explored by Bateman and bears its name, while the second one relies on matrix exponential.

2.1 Bateman equations and their shortcomings

2.2 Matrix exponential: break your brains

Now notice that system (1) can be re-written in matrix form. Just a simple change in notation, nothing more:

$$\frac{d}{dt}\begin{vmatrix} [^{99}Mo] \\ \\ [^{99m}Tc] \end{vmatrix} = \begin{pmatrix} -\lambda_{Mo} & 0 \\ \\ b\lambda_{Mo} & -\lambda_{Tc} \end{pmatrix} \begin{vmatrix} [^{99}Mo] \\ \\ [^{99m}Tc] \end{vmatrix} \tag{2}$$

Only the subdiagonal entry picks up the factor b. Diagonalizing exactly as before (eigenvalues still $-\lambda_1, -\lambda_2$ since those live on the diagonal regardless of b):

$$e^{At} = \begin{pmatrix} e^{-\lambda_1 t} & 0 \\[4pt] \dfrac{b\lambda_1}{\lambda_2-\lambda_1}\left(e^{-\lambda_1 t}-e^{-\lambda_2 t}\right) & e^{-\lambda_2 t} \end{pmatrix}$$

So Bateman becomes:

$$N_2(t) = N_1^0\,\frac{b\lambda_1}{\lambda_2-\lambda_1}\left(e^{-\lambda_1 t}-e^{-\lambda_2 t}\right)$$

— literally just the un-branched formula times b. This makes sense: branching ratio only ever multiplies the *production* term, never the decay-loss term, so it factors out linearly and leaves the transient/secular equilibrium math (the classic 6-hour Mo-99/Tc-99m generator "milking" curve) otherwise unchanged.

Where it stops being this simple

If you extend the chain further (e.g. tracking Tc-99 ground state buildup too, or handling isotopes with 3+ branches like your Ac-225 chain has at Bi-213 → Tl-209/Po-213), you get multiple daughters fed from one parent. That's not a single chain anymore — A becomes a genuine tree structure, and you'd want one column feeding multiple rows (each weighted by its own b_i, with $\sum b_i = 1$) rather than a simple bidiagonal matrix. The matrix exponential machinery still works fine, it's just no longer tridiagonal/bidiagonal, so `expm` is basically mandatory rather than a convenience at that point.

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- Data availability: The article and the app code are available on GitHub

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